

VERA X16 for Atari — Running the tests in the emulator

atari800 fork with the VeraX16 PBI card

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1 Overview

The test programs in **vera-tests/** run on a fork of the **atari800** emulator (version 5.2.0) that emulates the VeraX16 card on the Atari PBI bus: VERA registers at **\$D100-\$D11F**, the 2 KB handler ROM at **\$D800-\$DFFF** (selected through the PBI latch **\$D1FF**).

The emulator lives next to this repository:

```
~/Progetti/ATARI/
├─ atari800/                  emulator (src/atari800 is the binary)
│   └─ vera_pbi_rom -> ../VERA_ATARI_PBI
└─ VERA_ATARI_PBI/          this repository (ROM, drivers, tests)
```

All the commands below are run **from the root of this repository** (VERA_ATARI_PBI), so the emulator is **../atari800/src/atari800**. For brevity:

EMU=../atari800/src/atari800

2 Prerequisites

2.1 Building the emulator

```
cd ../atari800
./autogen.sh
./configure --enable-pbi-verax16
make
cd -
```

The VERA emulation is in `src/pbi_verax16.c` (registers, FX, audio, SPI) and `src/vera_video.c` (rendering). Without `--enable-pbi-verax16` the `-verax16` options do not exist.

2.2 Atari OS ROMs

If `atari800` was built with `libcurl` (`configure` enables the download feature on its own when it finds it) and no OS ROM file is configured, at start-up it **downloads the ROMs from the Internet**: it fetches <http://www.emulators.com/freefile/pcxf380.zip> (PC Xformer 3.8), extracts the `.rom` files into `../atari800/src/rom/`, uses them (original Atari XL OS and BASIC) and writes their paths into `../atari800/src/.atari800.cfg`. From then on it does not download them again.

`atari800` reads `.atari800.cfg` from the directory of the executable before `~/atari800.cfg`: once `../atari800/src/.atari800.cfg` exists, the emulator `../atari800/src/atari800` no longer reads `~/atari800.cfg`.

To use the built-in AltirraOS and Altirra BASIC, with no download, build with:

```
./configure --enable-pbi-verax16 --disable-download
```

and leave the `ROM_OS_*` paths empty in the configuration file.

2.3 Building the ROM, the drivers and the test disks

Required tools: `cc65` (`ca65`, `ld65`, `cl65`), `dir2atr`, `python3`.

```
make          # ROM, VERA*.SYS drivers, test programs, disk images
```

Main outputs:

File	Content
<code>vera_pbi_handler.rom</code>	PBI handler ROM, 2 KB (boots in 80×60)
<code>VERA4030.SYS</code> , <code>VERA8030.SYS</code> , <code>VERA8060.SYS</code>	RAM driver for 40×30, 80×30, 80×60
<code>disk1-runcpm.atr</code>	DOS 2.0S, <code>RUNCPM.COM</code> , <code>VERA8030.SYS</code>
<code>disk2-veratests-40x30.atr</code>	TEST4, TESTGS4, TESTMAZ4, TESTMTX4, TESTRMT, <code>VERA4030.SYS</code>
<code>disk2-veratests-80x30.atr</code>	TEST8, TESTGS8, TESTMAZ8, TESTMTX8, TESTRMT, <code>VERA8030.SYS</code>
<code>disk2-veratests-80x60.atr</code>	TEST6, TESTGS6, TESTMAZ6, TESTMTX6, TESTRMT, <code>VERA8060.SYS</code>
<code>disk3-standalone.atr</code>	TESTFX, TESTIRQ, TESTPLR + <code>DEM0.VTM</code> (tests without the driver)
<code>disk4-rmtio.atr</code>	TESTRIO with MyPicoDos autorun and the test files

The `TESTn/TESTGSn/TESTMAZn/TESTMTXn` programs on the `disk2` disks already contain the driver of their resolution ($n = 4: 40 \times 30, 8: 80 \times 30, 6: 80 \times 60$).

3 The basic command

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \
    disk3-standalone.atr
```

- `-verax16` plugs the card in; `-verax16-rom` gives the handler ROM.

- `-xl` (800XL) or `-xe` (130XE): the driver needs an XL/XE machine.
- `-pal` or `-ntsc`: the tests must pass in both.
- `-nopatch`: no high-speed SIO patch, disk I/O runs at the real serial speed (the H: device keeps working). To make it permanent, set `ENABLE_SIO_PATCH=0` in `~/atari800.cfg`.
- The last argument is the disk in `D1:`; more `.atr` files go to `D2:`, `D3:`...

The emulator opens two outputs: the normal ANTIC/GTIA screen and the VERA screen (640×480 VGA).

3.1 Starting a program from DOS 2.0S

The disks boot DOS 2.0S. From the DUP menu:

1. press `L` (BINARY LOAD);
2. type the file name, for example `TESTFX.COM`, and press Return.

3.2 Starting a program without a disk

`-run` loads an executable straight from the host file system:

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \
    -run TESTIRQ.COM
```

4 Options

4.1 VeraX16 card options

Option	Meaning
<code>-verax16</code>	enable the card (alias <code>--use-verax16</code>)
<code>-verax16-rom F</code>	handler ROM, 2 KB at <code>\$D800-\$DFFF</code>
<code>-verax16-pbi-id N</code>	PBI device bit 0-7 (default 7, mask <code>\$80</code>)
<code>-verax16-config-ms N</code>	FPGA configuration time after power-on, bus dead meanwhile (default 100); 0 = the board holds Atari RESET until <code>CONFIG_DONE</code>
<code>-verax16-debuglevel N</code>	log level (default 0)
<code>-verax16-psg-volume N</code>	VERA PSG level in percent, 0-400 (default 100 = one voice as loud as a POKEY channel at volume 15)
<code>-verax16-sdcard F</code>	raw SD image (e.g. from <code>dd</code>) exposed through the VERA SPI

4.2 Useful atari800 options

Option	Use
<code>-xl, -xe</code>	800XL / 130XE machine
<code>-pal, -ntsc</code>	video standard
<code>-basic, -nobasic</code>	BASIC ROM on/off
<code>-run F</code>	run a COM/EXE/XEX/BAS file
<code>-nopatch</code>	no high-speed SIO patch: real serial timing, H: still works
<code>-nopatchall</code>	no OS patches: real SIO serial timing (H: does not work)
<code>-volume N</code>	output volume 0-100
<code>-stereo</code>	two POKEYs
<code>-turbo</code>	as fast as possible
<code>-netsio [port]</code>	NetSIO for FujiNet-PC (default UDP 9997)
<code>-config F</code>	alternate configuration file

5 Running each test

5.1 TESTFX — FX coprocessor

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
disk3-standalone.atr
```

From DUP: L, TESTFX.COM. It checks every FX register and measures VRAM copy/fill throughput. Expected result: **PASS 36, FAIL 0**.

5.2 TESTIRQ — VERA interrupts

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
-run TESTIRQ.COM  
$EMU -xl -ntsc -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
-run TESTIRQ.COM
```

It checks the VMIRQ hook: VSYNC at 59.94 Hz, line IRQ, AFLOW masking, chaining to the OS and removal. Expected result: **PASS 13, FAIL 0**, both in PAL and in NTSC. See [Documentation/VERA-IRQ.md](#).

5.3 TESTPLR — VTM player

```
$EMU -xl -pal -nopatch -volume 100 -verax16 -verax16-rom vera_pbi_handler.rom \  
disk3-standalone.atr
```

From DUP: L, TESTPLR.COM. Use `-pal` or `-ntsc` according to the frame rate the song was converted for. `-volume 100` is needed: the VERA PSG is quiet at the default mixer level (or raise `-verax16-psg-volume`).

5.4 TEST / TESTGS / TESTMAZ / TESTMTX — driver and screen modes

```
$EMU -xl -pal -nopatch -verax16 -verax16-rom vera_pbi_handler.rom \  
disk2-veratests-80x30.atr
```

Use the disk of the resolution under test (40x30, 80x30, 80x60) and start, from DUP, TEST8.COM, TESTGS8.COM, TESTMAZ8.COM, TESTMTX8.COM (digit 4 or 6 on the other disks): font loading, gradient, scrolling, maze and matrix demos.

5.5 TESTRMT — RMT player on POKEY + VERA

On every `disk2-veratests-*.atr` disk, it does not need VERA.SYS. From DUP: L, TESTRMT.COM.

Key	Output
1	POKEY only
2	VERA only
3	both (default)
4	hybrid, RMT8 only: POKEY channels 1-4, VERA channels 5-8
S	VERA stereo on/off
ESC	stop and exit

5.6 TESTRIO — music during disk I/O

```
make disk4-rmtio.atr  
$EMU -xl -pal -nopatchall -nobasic \  
-verax16 -verax16-rom vera_pbi_handler.rom disk4-rmtio.atr
```

The disk starts TESTRIO.COM by itself (MyPicoDos autorun). `-nopatchall` is mandatory: without it the emulator intercepts SIO and there is no real serial timing. Expected (PAL, RBL loader): ~720 bytes/s, 0 errors, 0 lost ticks.

5.7 RUNCPM — ANSI terminal through FujiNet

Terminal 1, FujiNet-PC:

```
cd FujiNet/fujinet-pc-ATARI
./run-fujinet -c fnconfig.ini -s SD/
```

Wait for **### NetSIO stopped ###** in the log, then in terminal 2:

```
$EMU -xl -pal -nopatch -netsio -verax16 -verax16-rom vera_pbi_handler.rom
```

Details in README-fujinet.md.

6 Troubleshooting

Symptom	Cause / fix
ERROR: VeraX16 PBI card not found (\$D100 not responding).	-verax16 is missing: every test calls vera_require() at start
Three beeps from the console speaker at boot	the ROM did not see the VERA within ~0.7 s; check -verax16-config-ms
Unknown option -verax16	emulator built without --enable-pbi-verax16
No sound or very quiet VERA	add -volume 100 or raise -verax16-psg-volume
A file is on the disk but DOS does not list it, or “0 FREE SECTORS”	dir2atr bug; rebuild with make, which runs vera-tests/tools/fix_atr_vtoc.py
SIO errors or wrong speed in TESTRIO	-nopatchall is missing